JEE MATHEMATICS FORMULA VAULT

Complete Derivatives, Integrals & Spatial Vector Matrix Math

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1. Derivatives & Standard Limits

Trig Limit `lim (x->0) [sin(x) / x] = 1`

Exponential Limit `lim (x->0) [ (e^x - 1) / x ] = 1`

Logarithmic Limit `lim (x->0) [ ln(1+x) / x ] = 1`

Derivative of Inverse Sin `d/dx [ sin^-1(x) ] = 1 / sqrt(1 - x^2)`

Derivative of Inverse Tan `d/dx [ tan^-1(x) ] = 1 / (1 + x^2)`

Derivative of Inverse Sec `d/dx [ sec^-1(x) ] = 1 / (|x| * sqrt(x^2 - 1))`

2. Integration Techniques & Formulas

Integration by Parts `Integral [ u * dv ] = u * v - Integral [ v * du ]`

ILATE Priority: Select the term "u" according to: Inverse trigonometric, Logarithmic, Algebraic, Trigonometric, and Exponential functions.

Standard integral type A `Integral [ dx / (x^2 - a^2) ] = (1/(2*a)) * ln| (x-a)/(x+a) | + C`

Standard integral type B `Integral [ dx / (a^2 - x^2) ] = (1/(2*a)) * ln| (a+x)/(a-x) | + C`

Standard integral type C `Integral [ dx / (x^2 + a^2) ] = (1/a) * tan^-1(x/a) + C`

Standard integral type D `Integral [ dx / sqrt(a^2 - x^2) ] = sin^-1(x/a) + C`

Standard integral type E `Integral [ dx / sqrt(x^2 +/- a^2) ] = ln| x + sqrt(x^2 +/- a^2) | + C`

Standard integral type F `Integral [ sqrt(a^2 - x^2) dx ] = (x/2)*sqrt(a^2 - x^2) + (a^2/2)*sin^-1(x/a) + C`

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Part 2: Vector Spaces, Determinants, & Linear Equation Matrices

3. Vector Dot, Cross, & Triple Products

Vectors are represented in 3D coordinate space with orthogonal unit vectors i, j, k.

Vector Dot Product	`A . B = A_x*B_x + A_y*B_y + A_z*B_z = |A| * |B| * cos(theta)`
Vector Cross Product	`A x B = Determinant matrix of [ [i, j, k], [Ax, Ay, Az], [Bx, By, Bz] ]`
Cross Product Magnitude	`|A x B| = |A| * |B| * sin(theta)`
Vector Triple Product	`A x (B x C) = (A . C) * B - (A . B) * C`
Scalar Triple Product	`[A B C] = A . (B x C)   [Volume of parallelopiped]`

4. Matrices, Cramer Rule, & Inverse Algebra

Cramer Rule: Solves sets of simultaneous linear equations by checking ratios of determinant values.

Variables Solve Ratios	`x = D_x / D,   y = D_y / D,   z = D_z / D   [For D !== 0]`

1. If D !== 0: System has a unique solution (Consistent).
2. If D === 0 and at least one D_i !== 0: System has no solution (Inconsistent).
3. If D === 0 and all D_i === 0: System has infinitely many solutions.

Matrix Inverse Rule	`A^-1 = (1 / |A|) * adj(A)   [For non-singular |A| !== 0]`